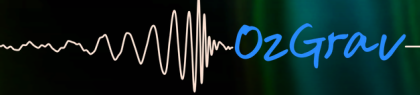


INVESTIGATING THE FORMATION OF HEAVY DOUBLE NEUTRON STARS

SWINBURNE

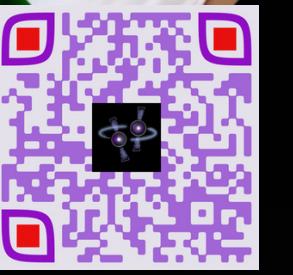
CENTRE FOR ASTROPHYSICS AND SUPERCOMPUTING

ARC Centre of Excellence for Gravitational Wave Discovery



Ashwathi Nair

Simon Stevenson

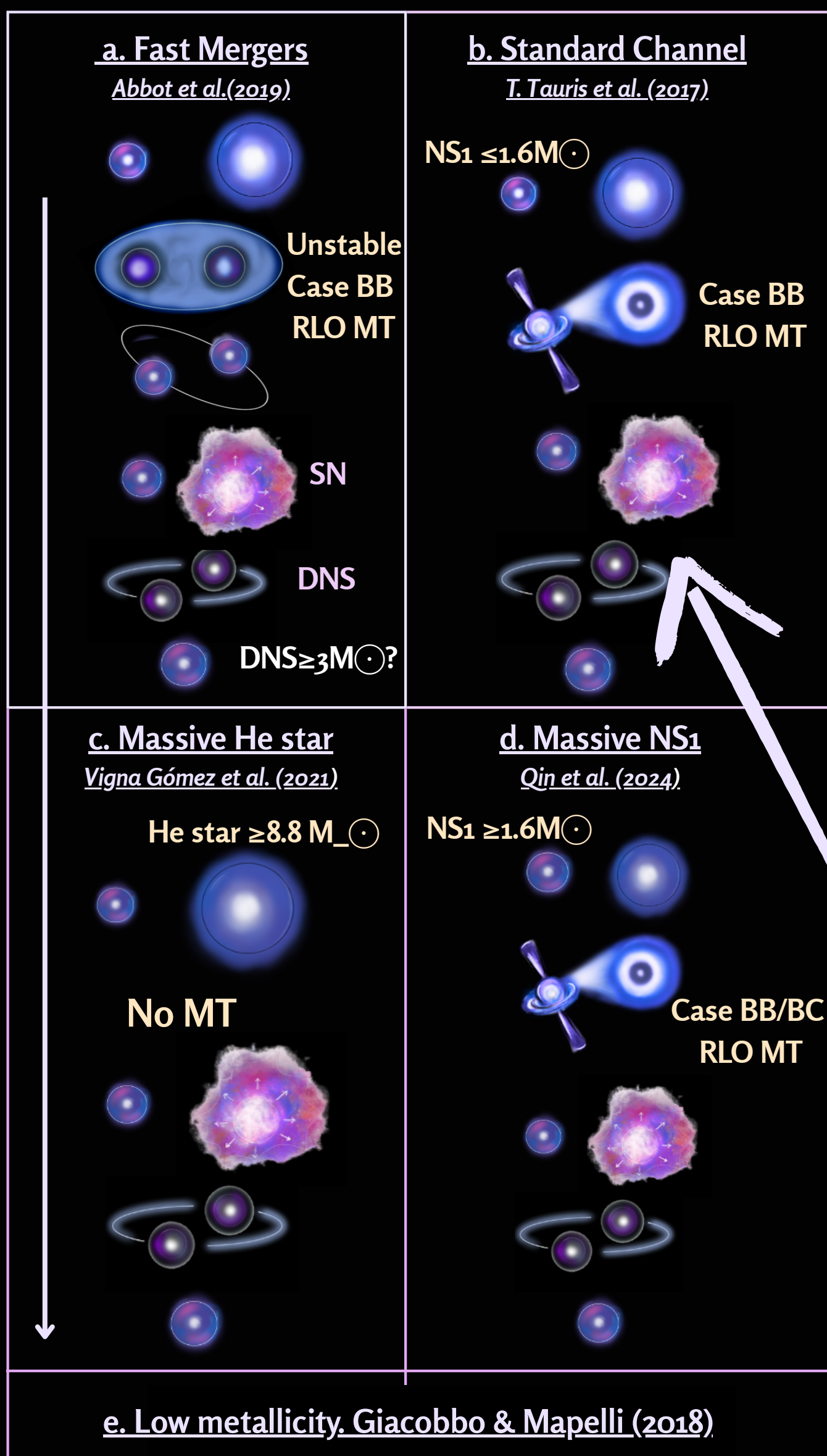


INTRODUCTION

Double Neutron Star (DNS) - GW190425 has a total mass of $3.4 M_{\odot}$, which is a bit massive compared to the observed DNS mass range in our Galactic disk. The unusual mass indicates that GW190425 might have formed differently than the observed Galactic DNS mergers and GW170817. A few alternate formation pathways have been proposed to explain this.

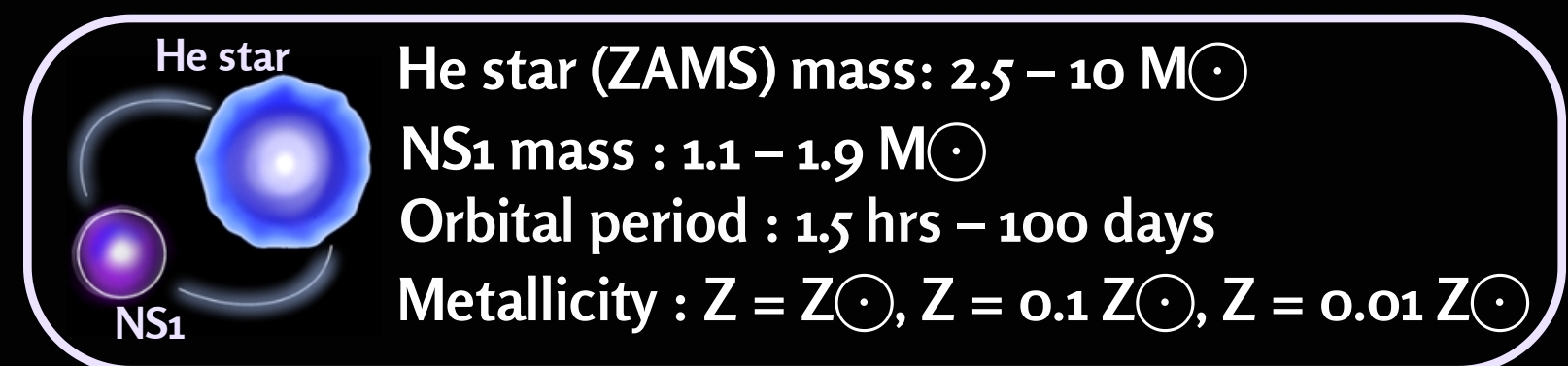
My role? To test these formation channels and determine which is more likely to give birth to such heavy DNSs, under a self consistent framework!

What are these alternate channels?



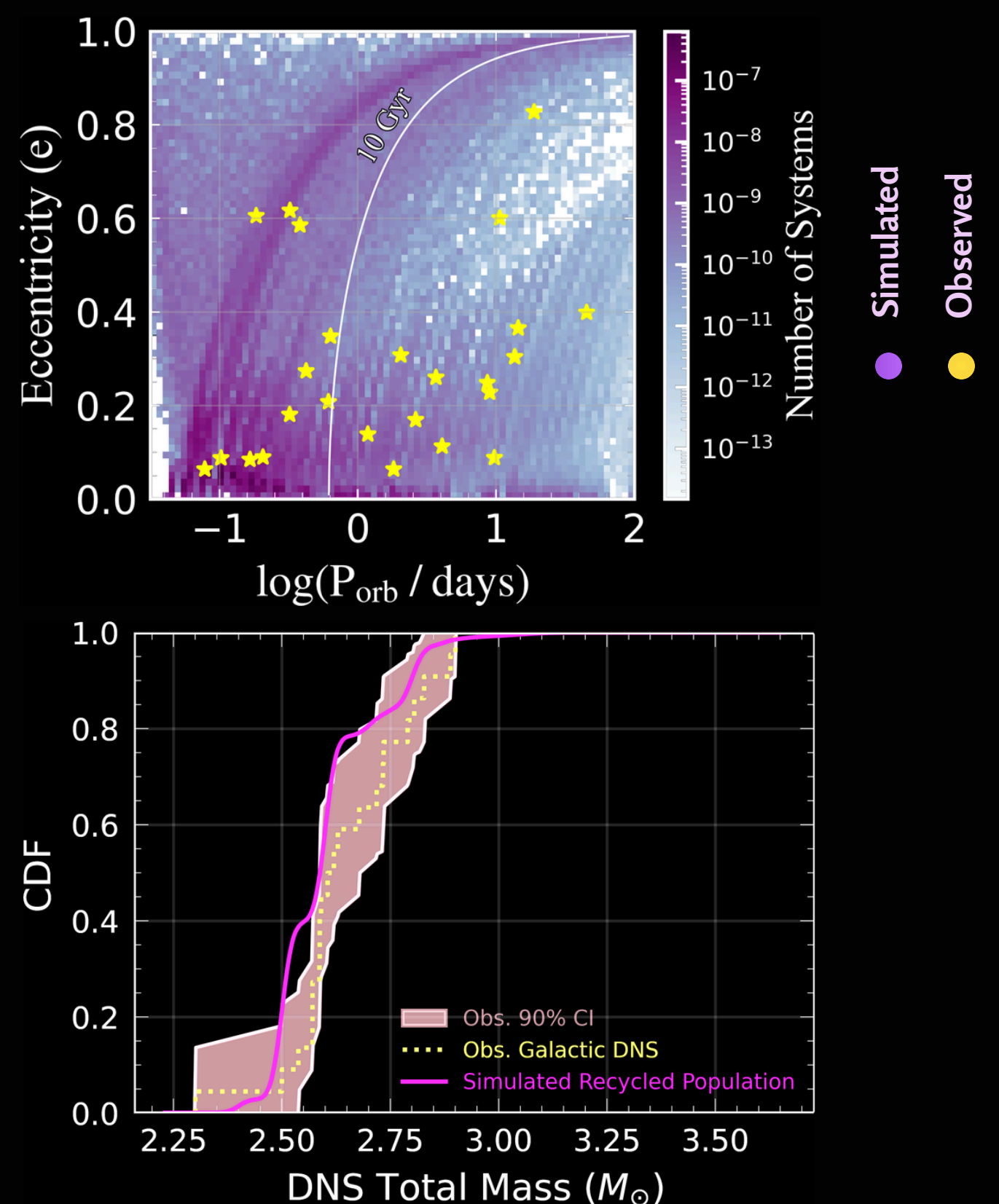
HOW DO WE TACKLE THIS?

By evolving ~ 6000 He star-NS binary models using 1D detailed stellar evolution code **MESA**



important sidequest before investigation : Can we explain the observed population of DNS in Galactic field ?

We can broadly!



AND WE HAVE A WINNER!!!

- Formation through the ‘Standard channel’ dominates at solar metallicity!
- We rule out ‘Fast Mergers’ at solar metallicity.
- Massive He star and Massive NS1 are rare in the Universe, they contribute insignificantly.
- Heavy DNS formation is **weakly dependent on metallicity**
- Formation of GW190425 like system is rare.